Modelling Pollinator Niches, Plant–Pollinator Coupling, and Community Turnover

Species Distribution Modelling (SDM) and Generalized Dissimilarity Modelling (GDM)

FELAB | Konkuk University (Data: National Ecosystem Survey 3rd–5th)

November 2026 | Seoul, South Korea

Introduction: From Species Niches to Ecological Interactions

- **Scope**: 2050 climate and land-use impacts across three ecological scales:
  - **Species**: Habitat suitability, realized thermal limits, and 2050 range shifts
  - **Interactions**: Spatial co-occurrence and mismatch with floral and host plants
  - **Communities**: High-elevation species replacement vs. lowland nested attrition
- **Workflow**: Presence records $\rightarrow$ 2050 projections $\rightarrow$ Plant–pollinator overlap $\rightarrow$ Stacked SDM (SSDM) + GDM

Step	Focus	Core Ecological Topic	Notebook Section & Slides
Step 1	**Niches & Drivers**	5 orders, 12 groups (**FG01–FG12**), 22 predictors	`Step 1–A / 1–B` (Slides 4–14)
Step 2	**SDM & Climate**	MaxEnt SDM, spatial block CV, & `Bio5` masks	`Step 2–A / 2–B` (Slides 15–26)
Step 3	**Habitat Coupling**	Plant–pollinator refugia & mismatch deficits	`Step 3` (Slides 27–31)
Step 4	**Beta Turnover**	Stacked SDM vs. GDM & β_{sim} / β_{sne} partitioning	`Step 4–A / 4–B` (Slides 32–41)

Hands-On Lab: Structure & Access Guide

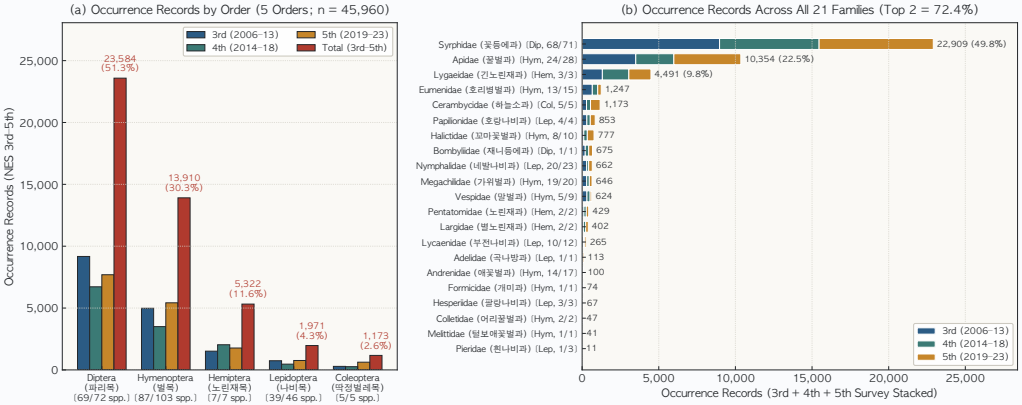
- **Interactive Notebook (`SDM_HandsOn_Workshop_KO.ipynb`)**:
  - **[실습] Single-Case**: Step-by-step exercise on **`FG03` *Bombus*** (`bioclim1` map, 50 km block CV, `Bio5` mask, *Bombus* × Spring Woody overlap, & 2-site Baselga β)
  - **[전체 비교] Synthesis**: `scripts/sdm_core.py` across **5 orders**, **12 groups (`FG01`–`FG12`)**, **8 plant–pollinator pairs**, and **4 community matrices**
- **How to Connect**:
  - **`MyBinder`**: **github.com/alanbseo/SDMworkshop** → **`Launch Binder`** badge *(direct: **mybinder.org/v2/gh/alanbseo/SDMworkshop/main**)*
  - **Local PC**: `git clone https://github.com/alanbseo/SDMworkshop.git` → `pip install -r requirements.txt` → `jupyter lab SDM_HandsOn_Workshop_KO.ipynb`

STEP 1. Ecological Niches & Environmental Drivers

- **Scope & Data Structure**
  - 5 insect orders and 12 functional groups (**FG01–FG12**) across 84,576 inland 1 km cells
  - Presence-only (PO) records and 10,000-cell presence–background (PB) calibration regarding imperfect detection
  - 22 climatic, water-balance, topo-soil, and 100 m land-cover predictors (2020 & 2050)
- **Objective**: Quantify how macroclimate and 100 m landscape structure differentiate pollinator niches.

[Step 1] Sample Distribution by Order and Family (NES 3rd–5th)

- **Sample Skewness (n = 45,960, 207 / 233 spp.)**: **Diptera (51.3%)** + **Hymenoptera (30.3%)** = **81.6%**, and **Syrphidae + Apidae** = **72.4%** across 21 families → motivates **12 balanced groups (`FG01`–`FG12`)**.



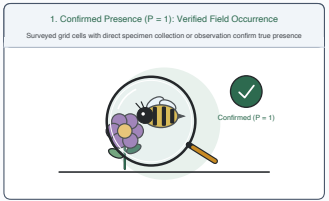
[Step 1] Pollinator Species and 12 Functional Groups (FG01–FG12)

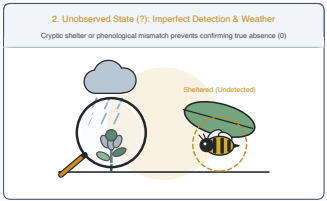
- **Grouping Criteria**: 12 groups defined by within-order sample size and ecological heterogeneity.

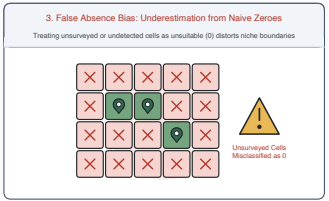
Order	Functional Group	Ecological Niche & Contrast	Cells
Hymenoptera (벌목)	**FG03** *Bombus* (뒤영벌속)	Cold-adapted montane bumblebees	927
Hymenoptera (벌목)	**FG04** *Apis* (꿀벌속)	Heat-tolerant eusocial honeybees	2,352
Hymenoptera (벌목)	**FG05** Other Apidae (기타 야생꿀벌과)	Carpenter & cuckoo wild bees	1,343
Hymenoptera (벌목)	**FG06** Solitary Bees (단독성 야생벌류)	Spring-active solitary wild bees	1,158
Diptera (파리목)	**FG01** Syrphidae (꽃등에과)	Widespread generalist hoverflies	2,753
Diptera (파리목)	**FG02** Bombyliidae (재니등에과)	Open-habitat parasitoid bee flies	519
Hemiptera (노린재목)	**FG07** Temp. Hemiptera (온대 노린재류)	Mainland flower & plant bugs	2,025
Hemiptera (노린재목)	**FG08** S. Largidae (큰별노린재과)	Subtropical coastal *Physopelta*	124
Lepidoptera (나비목)	**FG09** Papilionidae (호랑나비과)	Forest-interior swallowtails	535
Lepidoptera (나비목)	**FG10** Nymphalidae (네발나비과)	Forest-edge brush-footed butterflies	421
Lepidoptera (나비목)	**FG11** Other Lepidoptera (기타 나비·나방류)	Open/meadow butterflies & moths	296
Coleoptera (딱정벌레목)	**FG12** Cerambycidae (하늘소과)	Forest xylophagous flower longhorns	782

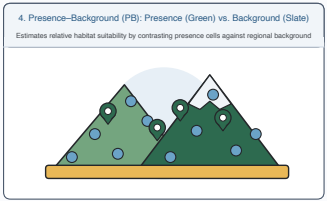
[Step 1] Imperfect Detection in National Survey Records

- **Unobserved** $\neq$ **Absent**: Survey gaps and weather create false absences (?), motivating presence–background (PB) modelling (Elith et al., 2011; Guillera-Arroita et al., 2015).



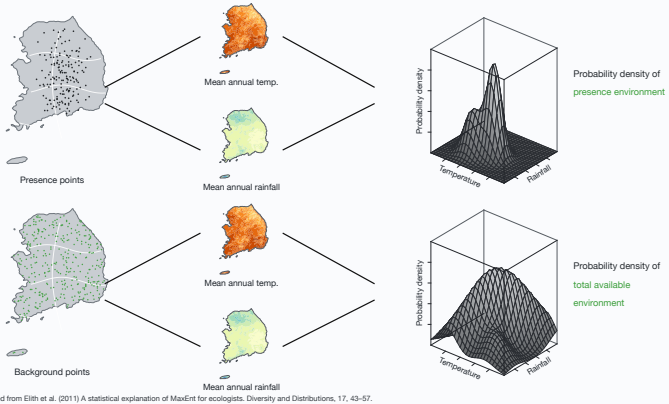






[Step 1] Presence–Background (PB) Niche Calibration (PO → MaxEnt)

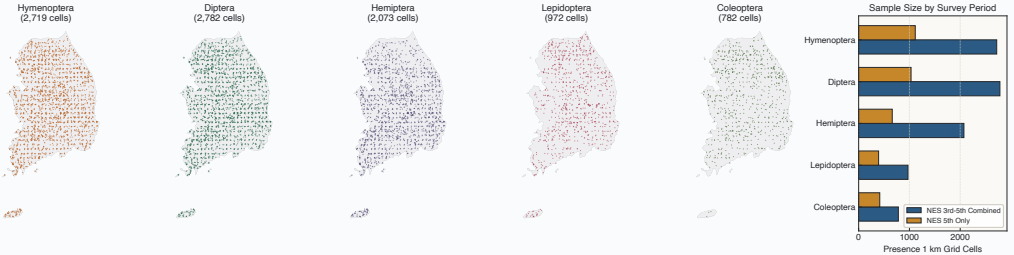
- **Density Ratio ($f_1(\mathbf{z})/f(\mathbf{z})$)**: Contrasts predictor densities at **presence sites** against the **1 km background** (Phillips & Dudík, 2008; Elith et al., 2011).



Adapted from Elith et al. (2011) A statistical explanation of MaxEnt for ecologists. Diversity and Distributions, 17, 43–57.

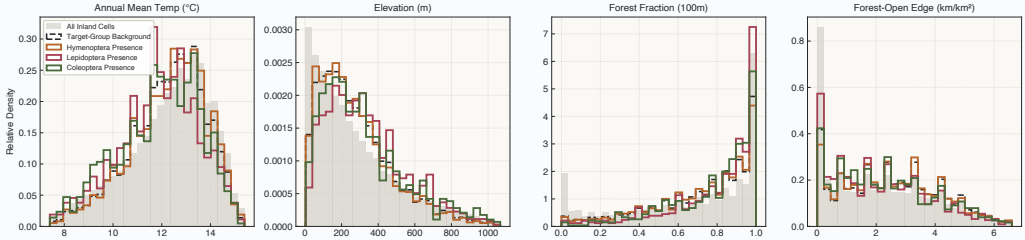
[Step 1] Biogeography and Survey Coverage Across Five Orders

- **Survey Coverage**: **4,523** surveyed 1 km cells (5.3% of 84,576 inland cells), led by Diptera (2,782) and Hymenoptera (2,719).



[Step 1] Forest-Edge Selection vs. Survey Sampling Bias

- **Survey Bias vs. Niche Selection**: Surveyed cells (`edge_F0`: 2.54 vs. 2.19 km/km² inland; Phillips et al., 2009) favor forest edges; **Hymenoptera** (2.56) and **Hemiptera** (2.62) select edge mosaics, whereas **Lepidoptera** (2.15) and **Coleoptera** (2.33) shift to closed forest (`frac_2`: 0.78–0.79).



[Step 1] Baseline Thermal and Habitat Niche Differentiation

- **Composite Habitat Profiles**: Standardized occupied-cell means separate cold-montane **FG03 *Bombus*** from generalist **FG04 *Apis*** and coastal **FG08**.

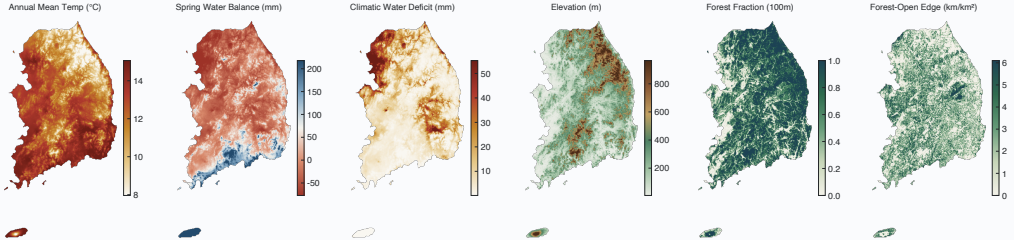
[Step 1] Input Data and Predictors: SDM vs. GDM

- **Shared vs. Specific Inputs**: Share **22 core 1 km rasters**, but differ in **biological response**, **masks**, and **spatial distance**.

Category	Variables / Codes	Model	Input Representation & Role
Response	Presence-only (PO) vs. Site × Spp.	**SDM** $\neq$ **GDM**	Presence–background (PB, y_i) vs. pair d_{ij}
Bioclimate (5)	`bioclim1, 2, 3, 13, 14`	**SDM & GDM**	Cell x_i (L/Q/H) vs. pair $\lvert\Delta I_k(x)\rvert$
Atmosphere (3)	`solarradiation, RH, VPD`	**SDM & GDM**	Solar radiation, humidity, vapor pressure deficit
Water Bal. (3)	`AI, WB_spr, CWD`	**SDM & GDM**	Aridity index, spring moisture, water deficit
Topo & Soil (5)	`elev, slope, soildeep3..stone`	**SDM & GDM**	Relief & soil depth/drainage/gravel
Land Cover (6)	`frac_0..3, edge_FO, shannon`	**SDM & GDM**	100 m cover, forest–open edge, diversity
Constraints	`bioclim5, 6, frac_6`	**SDM**	Occupied 99th/1st %ile (P_{99}/P_{01}) & water
Spatial	`Geo_Distance` (`x, y` km)	**GDM** (SDM CV)	GDM spline $I_{\text{geo}}(d_{ij})$; SDM 50 km CV

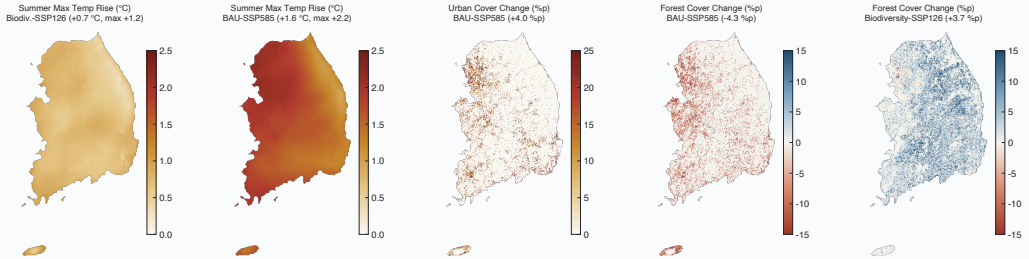
[Step 1] Baseline Environmental Gradients: Montane Refugia vs. Plains

- **2020 Baseline**: Cool, forested Baekdudaegan mountains vs. warm, drier lowland urban and cropland plains.



[Step 1] 2050 Climate and Land-Use Scenarios

- **Three Scenarios**: **BAU-SSP585** (+1.7 °C; `Bio5` +1.6 °C mean, max +2.2 °C; urban +4.0%p, forest -4.3%p) · **Climate-SSP126** (+0.6 °C; afforestation +3.4%p) · **Biodiversity-SSP126** (+0.6 °C; sprawl +0.3%p, forest +3.7%p).



STEP 2. Species Distribution & Climate Change

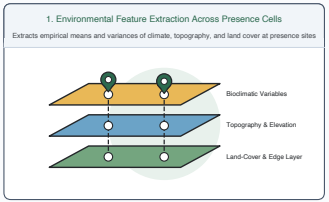
- **Calibration, Validation, & 2050 Projection**
  - Prevalence-balanced L1/L2 PB SDM on MaxEnt L/Q/H features ($\beta_{\text{reg}} = 1.5$; Phillips & Dudík, 2008; Renner & Warton, 2013)
  - **50 km × 50 km spatial block CV** (Roberts et al., 2017; AUC, TSS [Allouche et al., 2006]) & realized `Bio5` masks (Araújo et al., 2013)
  - 2050 habitat loss, gain, and elevational/latitudinal shifts across **FG01–FG12**

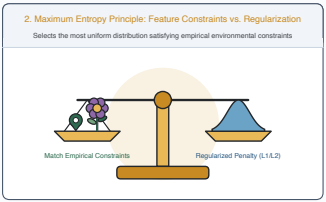
```
model = train_and_validate_sdm(
    pres_ids, env_2020, bg_grid, beta_reg=1.5, use_hinge=True, n_folds=3
)
```

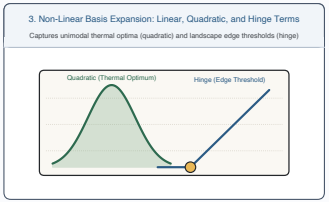
- **Objective**: Identify climate-vulnerable functional groups and persistent montane/coastal refugia.

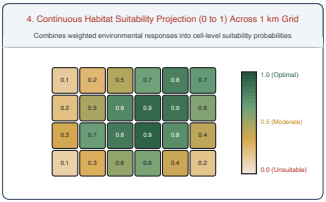
[Step 2] Capturing Thermal Optima and Habitat Thresholds

- **MaxEnt Basis Expansion**: Fits standardized Linear, Quadratic (thermal optima), and Hinge (edge thresholds) features under L1/L2 regularization (Phillips & Dudík, 2008).



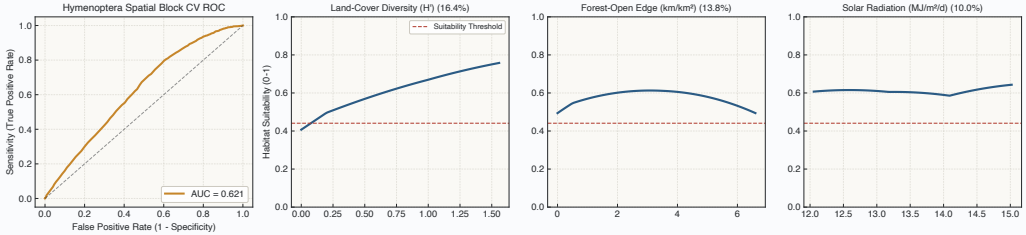






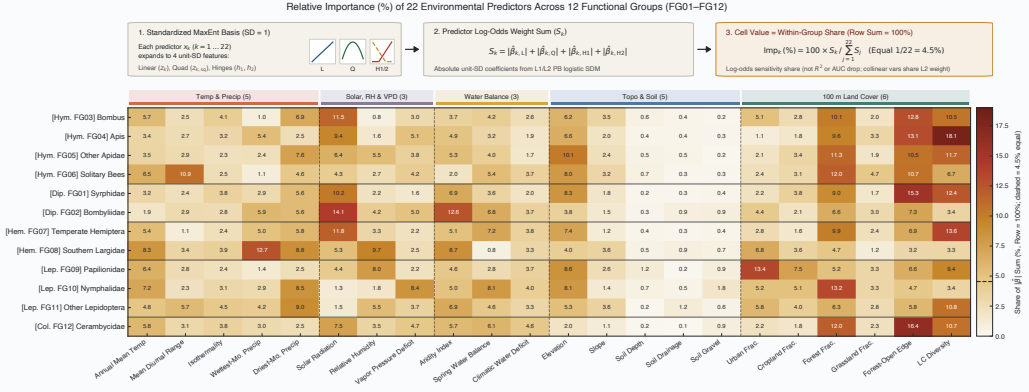
[Step 2] Response Curves: Edge & Mosaic Dependence in Bees

- **Hymenoptera Diagnostics**: Spatial CV **AUC = 0.621** (TSS = 0.195, τ = 0.441); standardized $|\hat{\beta}|$ share led by **Diversity (16.4%)**, **Edge (13.8%)**, and **Solar (10.0%)**.



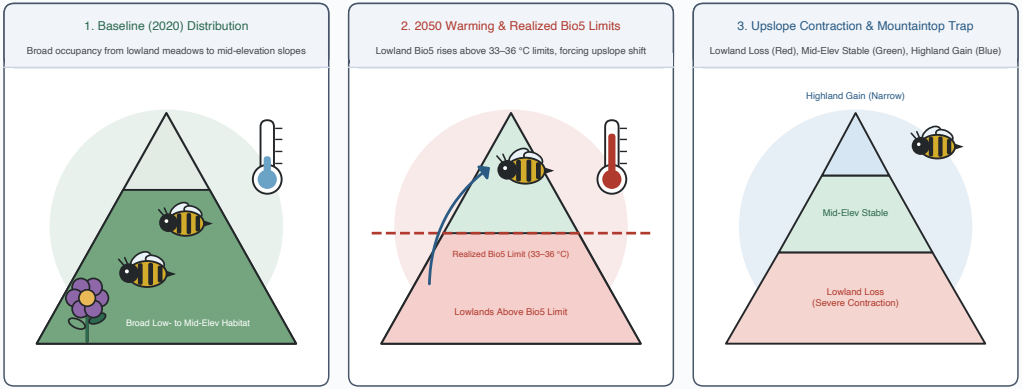
[Step 2] Variable Importance: Edge Mosaic vs. Forest Specialists

- **Metric**: Row share of unit-SD L/Q/H $|\hat{\beta}|$ sums (row $=$ 100%, equal = **4.5%**).
- **Group Controls (22 Predictors)**: Forest–open edge and diversity (edge_F0, shannon) lead **FG01 Syrphidae** (27.7%), **FG04 *Apis*** (31.2%), and **FG12** (27.1%); **FG10** tracks Forest Fraction (13.2%).



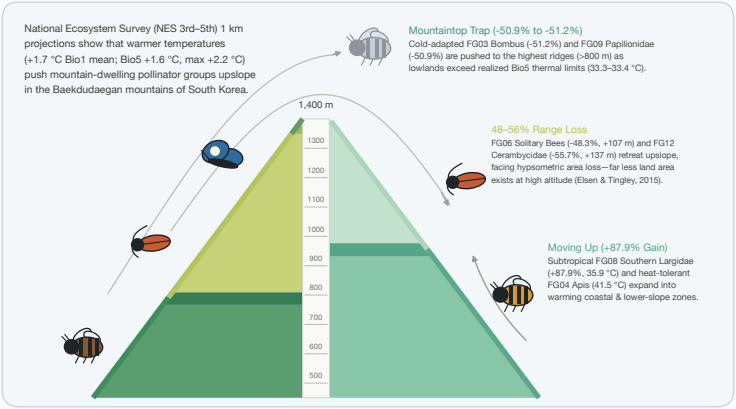
[Step 2] Lowland Heat Stress and Realized Thermal Limits

- **Summer Warming**: Where 2050 summer maxima (`Bio5`) exceed realized upper niche limits (33–36 °C; Araújo et al., 2013), lowland habitats contract upslope.



[Step 2] Lowland Loss, Upslope Contraction, and Mountaintop Traps

- **Hypsometric Mountaintop Trap**: Shrinking high-elevation area (Colwell et al., 2008; Elsen & Tingley, 2015) compresses cold-montane **FG03, FG09, FG12**, while **FG08** expands.



[Step 2] Realized Thermal Cutoffs: Global GBIF and National NES

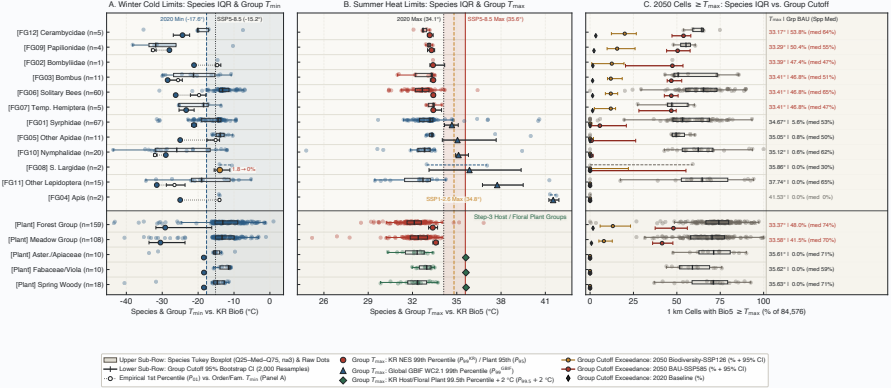
- **Rationale**: National 99th %iles cap warm niches at Korea's ceiling ($P_{99}^{\text{KR}} \leq 33.5$ °C), whereas **WorldClim 2.1** (`10'`, 1970–2000; Fick & Hijmans, 2017) is **3.30 °C cooler** than 1 km **KMA** (2021–2025) and truncates $> 5\%$ of Korean presences.

Cutoff Source	Occurrence & Climate Grid	Occupied-Cell Percentile Rule
1. Global (P_{99}^{GBIF}, P_{01}^{GBIF})	**GBIF** (233 poll./274 pl.) + **WC 2.1** (`10'`)	`10'`-thinned 99th/1st %ile of `Bio5`/`Bio6`
2. National (P_{99}^{KR}, P_{01}^{KR})	**NES 3rd–5th** + **KMA** (1 km, 2021–25)	1 km PO 99th/1st %ile (ΔBio5 = +3.30 °C)
3. Merged Cutoff	`Thermal_Limits.json` (Bio5/6)	$T_{\text{max}} = \max(P_{99}^{\text{KR}}, P_{99}^{\text{GBIF}})$; buffered T_{min} (FG08: P_{01})

- **Effect**: Separates cold-montane **FG03 *Bombus*** (**33.41 °C**) from warm-tolerant **FG08** (**35.86 °C**) and **FG04 *Apis*** (**41.53 °C**); realized macroclimatic bounds ($\neq$ physiological CT_{max}; Araújo et al., 2013).

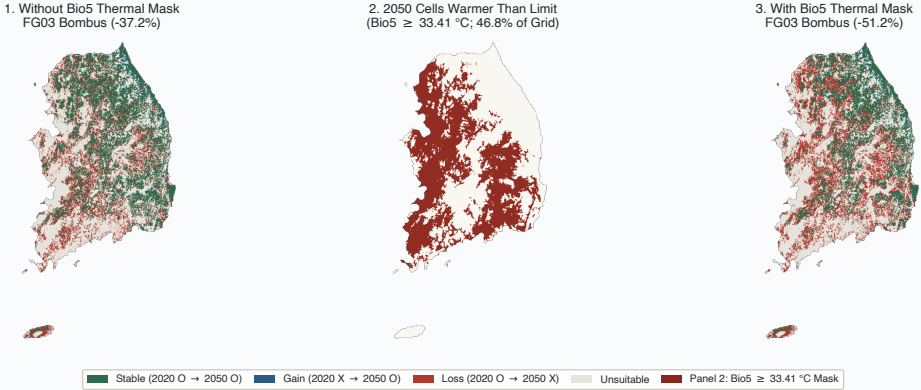
[Step 2] Thermal Cutoffs

- **BAU-SSP585 `Bio5` (34.1 → 35.6 °C)**: Group cutoffs (T_{max}) exclude **46.8–53.8%** of cells in cool-adapted groups (**FG02, 03, 06, 07, 09, 12**) vs. **0.0–5.6%** in warm-tolerant groups (**FG01, 05, 10, 11**).



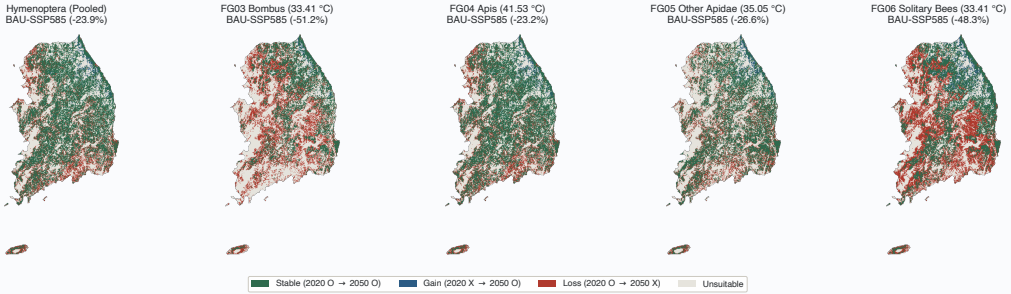
[Step 2] Realized Upper Thermal Masks Expose Lowland Loss

- **Climatic Niche Cutoff (FG03 *Bombus*, `Bio5` $\geq$ 33.41 °C)**: Applying realized upper thermal masks sharpens BAU-SSP585 loss from -37.2% to **-51.2%**.



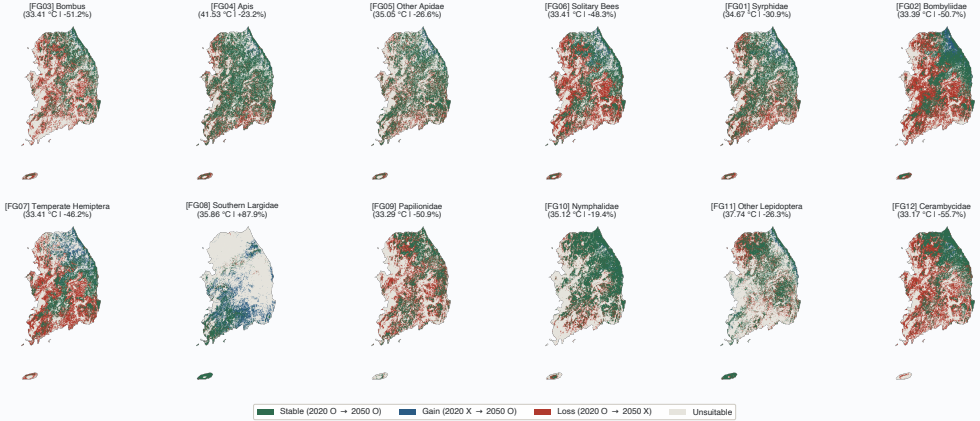
[Step 2] Contrasting Responses Within Hymenoptera (FG03–FG06)

- Hymenoptera (-23.9%, BAU-SSP585) masks losses in **FG03 *Bombus* (-51.2%)** and **FG06 Solitary (-48.3%)** vs. **FG04 *Apis* (-23.2%)**.



[Step 2] Range Shifts Across All 12 Functional Groups (FG01–FG12)

- Cool-adapted groups (**FG02, FG03, FG06, FG07, FG09, FG12**) lose **46–56%**, whereas subtropical **FG08** expands **+87.9%**.



[Step 2] Quantitative Summary of 2050 Range Shifts (FG01–FG12)

- **Shift Metrics (BAU-SSP585)**: $\Delta\bar{z}$ = mean elevation shift (72–100% from lowland loss); $\Delta\bar{y}$ = northward centroid shift (**FG08**: -1.2 m, **+24.3 km**).

Group	Order	Cells	AUC	$T_{\max}$	2020 Area	SSP126	SSP585	Elev. $\Delta\bar{z}$	North $\Delta\bar{y}$
FG03 *Bombus*	Hym.	927	0.628	33.4 °C	43,201 km²	-14.9%	**-51.2%**	+80.9 m	+31.3 km
FG04 *Apis*	Hym.	2,352	0.625	41.5 °C	54,100 km²	-6.3%	-23.2%	+39.5 m	+16.7 km
FG05 Other Apidae	Hym.	1,343	0.641	35.1 °C	45,603 km²	-22.5%	-26.6%	+32.4 m	+16.1 km
FG06 Solitary Bees	Hym.	1,158	0.623	33.4 °C	53,338 km²	-23.1%	**-48.3%**	+106.5 m	+33.8 km
FG01 Syrphidae	Dip.	2,753	0.620	34.7 °C	50,667 km²	-6.2%	-30.9%	+57.5 m	+26.6 km
FG02 Bombyliidae	Dip.	519	0.616	33.4 °C	57,951 km²	-18.9%	**-50.7%**	+139.1 m	+38.3 km
FG07 Temp. Hemiptera	Hem.	2,025	0.621	33.4 °C	47,514 km²	-41.5%	**-46.2%**	+127.4 m	+49.9 km
FG08 S. Largidae	Hem.	124	0.777	35.9 °C	12,660 km²	+128.3%	**+87.9%**	-1.2 m	**+24.3 km**
FG09 Papilionidae	Lep.	535	0.644	33.3 °C	46,597 km²	-18.1%	**-50.9%**	+67.6 m	+17.8 km
FG10 Nymphalidae	Lep.	421	0.680	35.1 °C	46,278 km²	+12.6%	-19.4%	+33.4 m	+17.9 km
FG11 Other Lepidoptera	Lep.	296	0.620	37.7 °C	36,237 km²	+38.2%	-26.3%	+15.7 m	+1.1 km
FG12 Cerambycidae	Col.	782	0.642	33.2 °C	53,089 km²	-20.3%	**-55.7%**	+136.5 m	+48.2 km

STEP 3. Plant–Pollinator Coupling & Spatial Mismatch

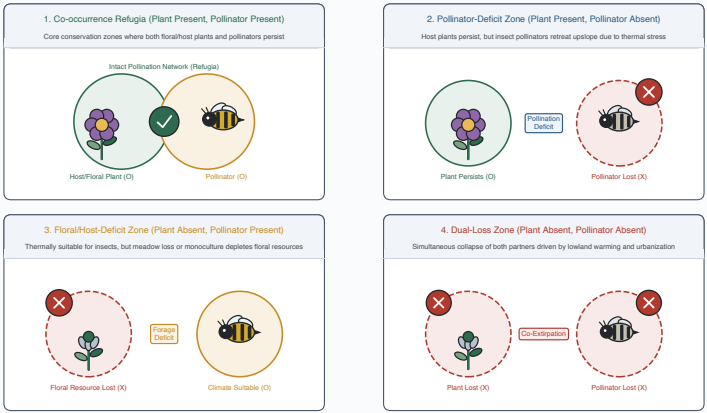
- **Biotic Decoupling Under Climate Change**:
  - Divergent range shifts erode spatial co-occurrence even in generalist networks (**Schweiger et al., 2008; Schleuning et al., 2016; Briggs & Anderson, 2026**); abiotic SDM risk alone poorly predicts secondary extinction risk ($r < 0.3$ under phenological mismatch; **Peng et al., 2025**)

Regime	Pollinator	Plant	Ecological Consequence
Co-occurrence refugia	Suitable	Suitable	Shared climatic & floral persistence
Pollinator deficit	Unsuitable	Suitable	Plants persist; pollinators lost to heat
Plant deficit	Suitable	Unsuitable	Climate suitable; host/floral plants lost
Dual loss	Unsuitable	Unsuitable	Simultaneous lowland co-extirpation

- **Objective**: Evaluate whether suitable insect climate space retains floral and host-plant overlap by 2050.

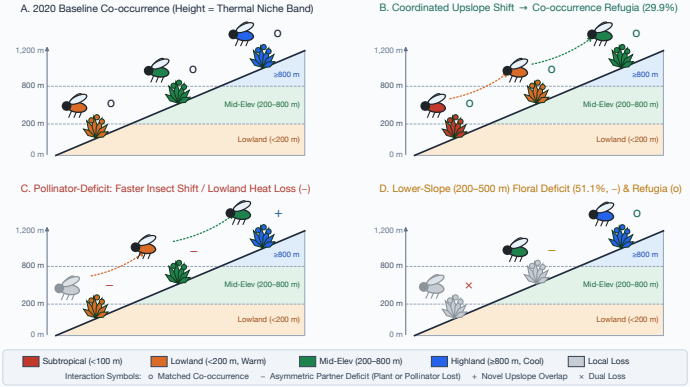
[Step 3] Spatial Decoupling of Pollinators and Host/Floral Plants

- **Overlay**: Classifies 1 km cells into **Co-occurrence Refugia**, **Pollinator Deficit**, **Floral/Host Deficit**, and **Dual Loss** (cf. Schweiger et al., 2008; Schleuning et al., 2016; Briggs & Anderson, 2026).



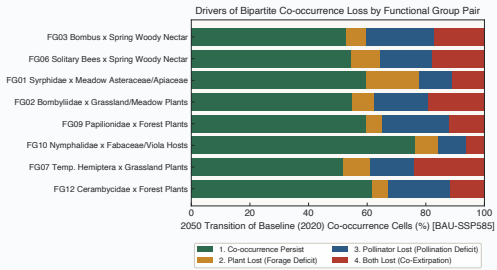
[Step 3] Spatial Mismatch Regimes Along Elevational Gradients

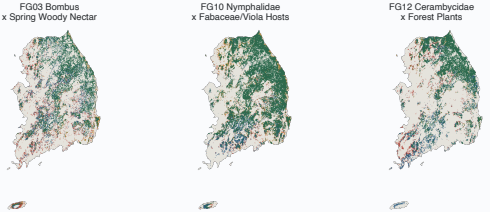
- Across lowland (<200 m), mid-slope (200–800 m), and highland ($\geq$ 800 m) bands, divergent range shifts (**Briggs & Anderson, 2026**) and host/floral loss (**Schweiger et al., 2008; Schleuning et al., 2016**) split baseline pairs (o) into deficits (–), novel overlap (+), and dual loss (×).



[Step 3] Contraction of Shared Pollinator–Plant Suitable Habitats

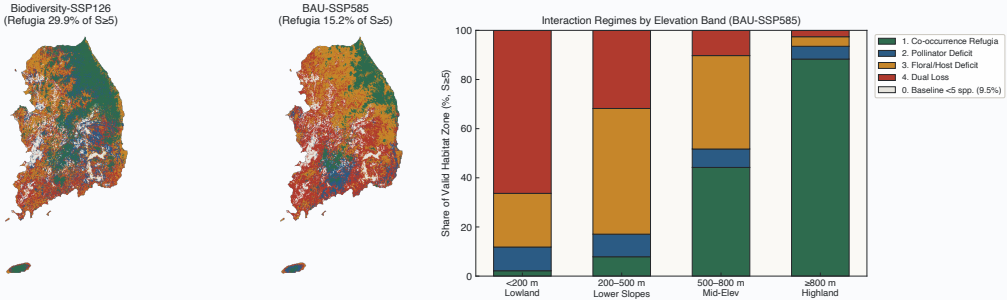
- **BAU-SSP585 Overlap (Schoener's D; Warren et al., 2008)**: Meadow loss (-23.9%) deepens **FG01 × Meadow** contraction (**-34.2%** vs. -30.9%); in **FG03 × Spring Woody** (**-43.0%**, D: 0.84 $\rightarrow$ 0.64), **23.1%** of 2020 shared cells lose *Bombus* (54.0% of 2050 plant cells lack *Bombus*).





[Step 3] Lower-Slope Floral Deficits and Co-occurrence Refugia

- **Community Synthesis (274 Plants × 139 Pollinators; $S_{2020} \geq 5$):** **Biodiversity-SSP126** retains **29.9%** national refugia ($\Delta S \geq$ -20%) vs. **15.2%** in **BAU-SSP585** (**51.1%** floral/host deficit at **200–500 m**).



STEP 4. Community Turnover & Ecological Synthesis

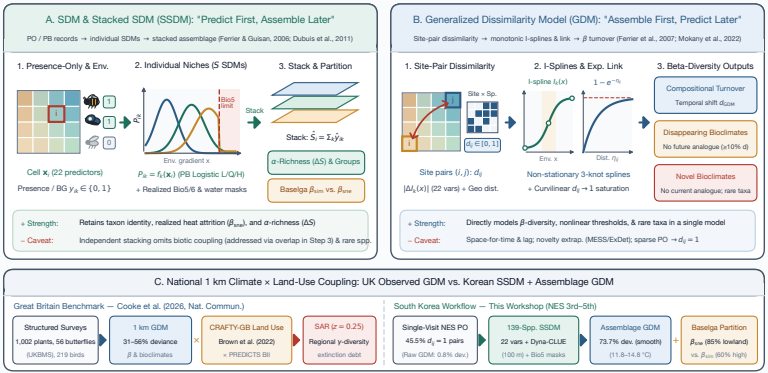
- **Framework Integration**
  - Integrates **SSDM** (Dubuis et al., 2011; Calabrese et al., 2014), **GDM** (Ferrier et al., 2007; Mokany et al., 2022), and **Baselga (2010)** partitioning ($\beta_{\text{sor}} = \beta_{\text{sim}} + \beta_{\text{sne}}$)

Dimension	Stacked SDM (SSDM)	Generalized Dissimilarity Modelling (GDM)
Main focus	Species distributions & richness (ΔS)	Nonlinear compositional turnover (d_{ij})
Response input	Single-cell PO/PB (y_{ik}) $\rightarrow$ suitability $p_{ik} \in [0, 1]$	Site-pair dissimilarity ($d_{ij} \in [0, 1]$, 25k pairs)
Predictors	22 env. vars (x_i) + `Bio5/6` & water masks	22 env. $\lvert\Delta I_k(x)\rvert$ + `Geo_Distance` spline
Main output	Richness & β_{sim} / β_{sne} maps	Monotonic I-spline turnover ($1 - e^{-\eta_{ij}}$)
Ecological role	Quantifies richness loss & nested attrition	Identifies critical turnover thresholds

- **Objective**: Partition 2050 compositional change into lowland nestedness loss vs. highland species replacement.

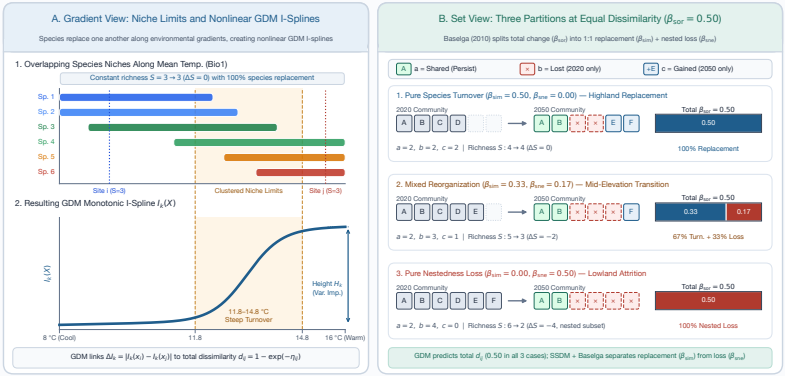
[Step 4] Conceptual Comparison: SDM (Species Stacking) vs. GDM

- **Paradigms**: **SDM / SSDM** (*predict first, assemble later*; Ferrier & Guisan, 2006; Zurell et al., 2020) vs. **GDM** (*assemble first, predict later*; Ferrier et al., 2007).
- **National Coupling**: **Cooke et al. (2026, *Nat. Commun.*)** link UK 1 km GDMs with **CRAFTY-GB** (Brown et al., 2022); we couple **Dyna-CLUE (100 m)** with **SSDM + assemblage GDM**.



[Step 4] Compositional Turnover: Gradient and Set-Partitioning Views

- **Gradient vs. Set Decomposition**: Clustered niche limits (**A**) produce steep nonlinear **GDM I-splines** even at constant richness ($\Delta S = 0$); equal total dissimilarity ($\beta_{sor} = 0.50$, **B**) spans **100% replacement** (β_{sim}) to **100% nested loss** (β_{sne}; Baselga, 2010).



[Step 4] Meaning of Baselga (2010) Beta-Diversity Partitioning

- **Decomposition**: Total Sørensen dissimilarity (β_{sor}) conflates **1:1 species replacement** ($\Delta S = 0$) with **uncompensated species loss** ($\Delta S < 0$).
- **Cell Counts (2020 $\rightarrow$ 2050)**: Shared species a (persist), lost species b (2020 only), gained species c (2050 only).

Component	Formula	Ecological Meaning & Limiting Case
Total (β_{sor})	$\frac{b+c}{2a+b+c} = \beta_{\text{sim}} + \beta_{\text{sne}}$	Overall Sørensen dissimilarity (0: identical, 1: disjoint)
Turnover (β_{sim})	$\frac{\min(b,c)}{a+\min(b,c)}$	**Species replacement** (Simpson): 1:1 substitution (0 if $b = 0$ or $c = 0$)
Nestedness (β_{sne})	$\beta_{\text{sor}} - \beta_{\text{sim}}$	**Nested subset loss**: attrition without replacement ($= \beta_{\text{sor}}$ if $c = 0$)

- **Role vs. GDM**: **GDM** models total d_{ij} only, whereas **SSDM + Baselga** separates highland **replacement** from lowland **attrition** (cf. β_{rich}; Carvalho et al., 2012).

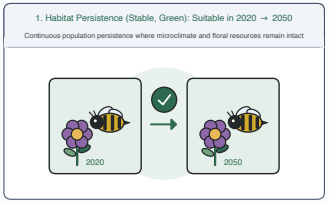
[Step 4] Methodological Assumptions & Extrapolation Limits

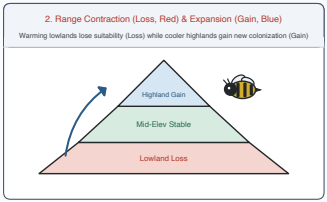
- **Correlative Projection Caveats**: **SSDM** and **GDM** project *potential* suitability rather than realized communities, omitting abundance, competition, and phenology (Blois et al., 2013; Peng et al., 2025; Briggs & Anderson, 2026).

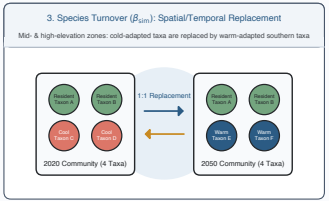
Assumption / Issue	Limitation in SDM / GDM	Diagnostic / Mitigation
Space-for-time	Assumes equilibrium & unlimited dispersal	Potential $\neq$ realized (Fitzpatrick & Keller, 2015)
Biotic & timing	Omits trophic links & phenology	Overlap (Briggs & Anderson, 2026)
Sparse PO	Sparse PO $\rightarrow d_{ij} = 1$; SSDM smooths	Raw PO (0.8%) vs. plant (37%) & UK (Cooke et al., 2026)
Novel climates	Extrapolation past baseline bounds	`Bio5/6` masks + **MESS** / **ExDet** (Mesgaran et al., 2014)

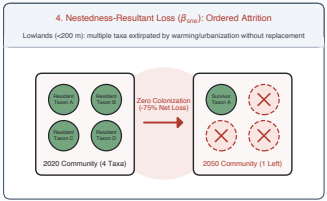
[Step 4] Single-Taxon Range Shifts vs. Community Beta Partitioning

- **Species vs. Community (Baselga, 2010)**: Connects taxon **Stable / Loss / Gain** to highland **Species Turnover (β_{sim})** and lowland **Nestedness-Resultant Loss (β_{sne})**.



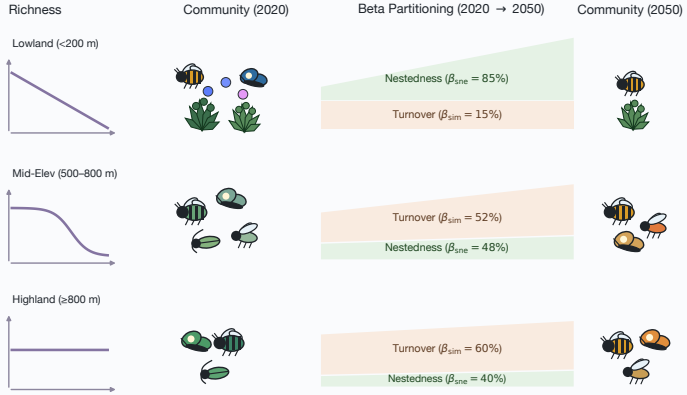






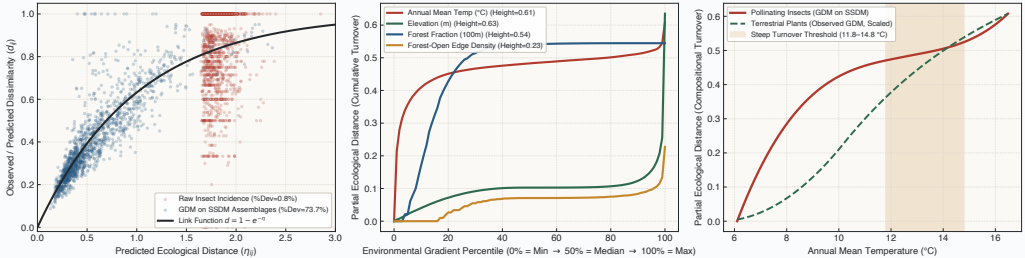
[Step 4] Richness Decline, Nested Attrition, and Species Replacement

- **Elevational Gradient (SSDM + Baselga, 2010)**: Lowland (<200 m) **nestedness loss** (β_{sne}: **85.2%**) $\rightarrow$ mid-elevation (500–800 m) transition $\rightarrow$ highland ($\geq$ 800 m) **replacement** (β_{sim}: **59.7%**).



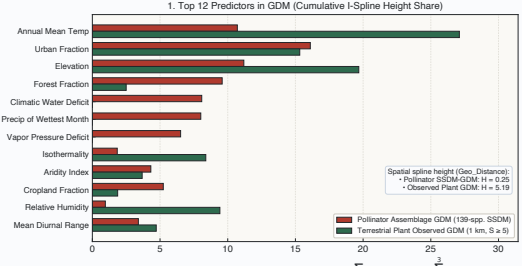
[Step 4] Nonlinear Thermal Turnover Thresholds (GDM I-Splines)

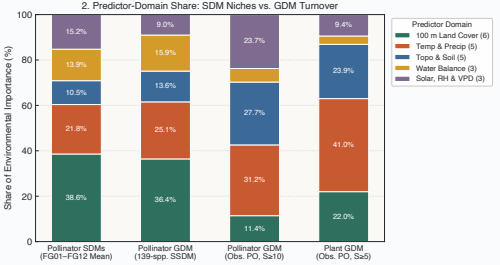
- **Survey Completeness**: Sparse single-visit insect PO saturates ($d_{ij} = 1$: 45.5%; **0.81%** dev.) vs. observed plants (**37.04%**) and UK transects (31–56%; Cooke et al., 2026); **139-spp. SSDM-GDM** captures smoothed turnover (**73.70%**, steepest at **11.8–14.8 °C**).



[Step 4] SDM Weights vs. GDM Spline Heights

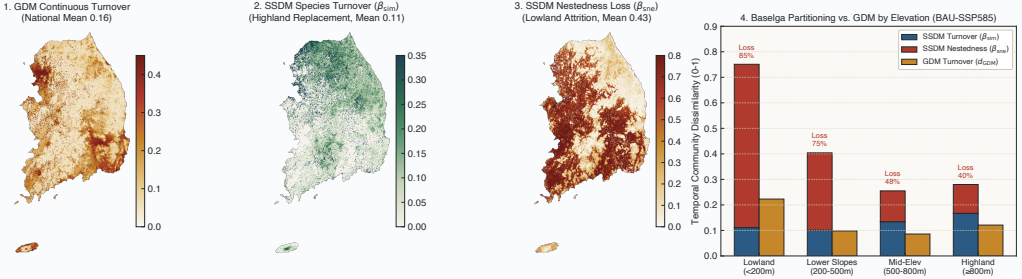
- **Importance Metrics**: **SDM** sums standardized $|\hat{\beta}|$ weights (land cover: **38.6%**), whereas **GDM** sums I-spline heights $H_k = \sum_{m=1}^{3} a_{km}$ (Ferrier et al., 2007; **36.4%** land cover in SSDM-GDM vs. **41.0%** climate in observed plants).





[Step 4] Lowland Community Attrition vs. Highland Replacement

- **Baselga Partitioning (BAU-SSP585)**: Lowlands (<200 m) are dominated by **nestedness-resultant loss** (β_{sne} = 0.640; **85.2%**), whereas highlands ($\geq$ 800 m) show **species replacement** (β_{sim} = 0.167; **59.7%**).



Summary

1. **Functional differentiation**
   - Group-level SDMs expose severe contractions masked by order pooling (**FG03 -51.2%**, **FG06 -48.3%**, **FG09 -50.9%**, **FG12 -55.7%**).
2. **Trophic & spatial connectivity**
   - Climate mitigation and habitat conservation (**Biodiversity-SSP126**) nearly double spatial co-occurrence refugia (**29.9%** vs. **15.2%** in **BAU-SSP585**), addressing interaction risks beyond single-taxon climate SDMs (Peng et al., 2025; Briggs & Anderson, 2026).
3. **Multidimensional community turnover**
   - **SSDM + GDM** (11.8–14.8 °C threshold) separates lowland nestedness attrition (β_{sne}: **85.2%**) from highland replacement (β_{sim}: **59.7%**).

- **Takeaway**: **Conserving pollinators requires more than climate space** — it requires protecting forest–open mosaics, host/floral plants, and spatial connectivity.

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